

Wireless Power Transfer Using a Cavity-Enclosed Impedance Metawire Structure

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Abstract—An impedance metawire structure consisting of printed circuit board (PCB) capacitor elements is presented for 10 GHz wireless power transfer (WPT) inside an enclosed conducting cavity for maximum power transfer efficiency (PTE). The design is a symmetric 40-wire, 6λ wide structure excited with a two element series-fed patch array, implemented on two layers of RO3003 substrate. This structure is then optimized for maximum PTE using full-wave simulation in COMSOL Multiphysics, targeting a resonant patch receiver positioned 1λ away. The results show high PTE of $\sim 85\%$ despite the system not being intentionally designed to exploit cavity resonances, with effective suppression of higher-intensity E fields (i.e. ≥ 100 V/m) in non-target regions compared to benchmark examples. The proposed cavity-enclosed metawire structure can be used for charging medical implants, securely charging devices, and general-purpose charging in existing enclosed metallic environments where cavity resonances cannot be fully exploited.

I. INTRODUCTION

Electromagnetic metawire structures are periodic or quasi-periodic arrangements of thin conductive or dielectric wires engineered to support prescribed electric and/or magnetic line currents under excitation, allowing control of scattering, radiation, and wave guidance. Compared to other types of metamaterials, metawire structures have the potential to be physically simpler, lower loss, and more easily fabricated. Depending on the wire spacing and design, metawire arrays can have many functions such as artificial impedance surfaces, leaky-wave radiators, diffraction gratings, or wavefront-shaping structures.

Impedance-loaded metawire structures are a specific implementation with lumped or distributed impedances wires periodically loaded along the wires, including capacitive, inductive, resistive, or active elements. For passive structures, which are the focus of this work, impedances can be readily realized in PCB form. In the context of WPT, metawire structures offer a compact and low-loss means of shaping electromagnetic fields, improving directionality, impedance matching, and near-field coupling efficiency. This work presents an application for enhancing WPT in a general-purpose scenario with a cavity-enclosed structure. The addition of a PCB metasurface not only enhances PTE, but also reduces off-target field amplitude, and does not rely on cavity resonances unlikely typical cavity WPT.

A. Wireless Power Transfer within Cavity Structures

A pair of T_x and R_x antennas in a fully enclosed perfect electric conductor (PEC) cavity can achieve very high PTE as radiated power cannot leave the system. Essentially 100% PTE is possible under ideal conditions, if the cavity is appropriately

sized and the antennas are placed at field maxima locations of cavity nodes. Generally, WPT applications in cavities will make use of such resonance modes [1], [2]. However for real-life applications it is not always possible or convenient to have the antennas positioned and the cavity sized in such a way that this effect can be fully utilized. Additionally, cutoff frequency constraints may not be satisfied, or the design frequency may be between modes, or the cavity may be severely over-/under-moded leading to chaotic fields and high sensitivity, or resonances may not be supported for various other reasons. This may be the case if the goal is to maximize PTE of two antennas already within an enclosed structure of fixed size (e.g. spacecraft/vehicle compartment). Furthermore, using such resonance modes typically results in high-field regions filling a greater internal volume fraction of the cavity which can cause unwanted inference if other components are present nearby, or for biomedical applications where fields in non-target regions should be generally minimized for health considerations. In this work a low-profile, passive metawire structure is presented which aims to address both issues: maximizing PTE within a PEC cavity, while achieving the secondary effect of reducing field intensity in non-target regions (i.e. away from the R_x antenna).

II. DESIGN

The proposed metawire structure is designed for 10 GHz, a widely used Industrial, Scientific, Medical (ISM) band for wireless power transfer and small electronic systems. The unit cells of the design consist of PCB capacitor elements (fig. 1), which are periodically distributed along each metawire (fig. 2). These capacitors are spaced by $\lambda/8$ along the wire axis, with their arm lengths adjusted to realize the required impedance. All capacitors on a given wire are identical, resulting in a structure that is invariant along the wire axis. The separation between adjacent metawires should remain below $\lambda/2$ to ensure a highly sub-wavelength configuration.

In this work the studied metawire structure uses $h_{\text{sub}} = 1.52$ mm (i.e. 60 mil) thick Rogers RO3003 substrate ($\epsilon_r = 3.0$). The specific choice of substrate permittivity is not critical; while RO3003 yields satisfactory performance, other substrates with comparable permittivity and loss tangent can also be employed. A top layer of $6\lambda \times 2\lambda$ (x-width, y-length, fig. 2) substrate with no PEC backing supports 40 metawires uniformly spaced 0.15λ apart. The wire count and spacing in this work were chosen for demonstrative purposes to ensure a small structure size, although other values are also viable. A second PEC-backed substrate layer is placed against the bottom surface of the cavity, $\lambda/8 - h_{\text{sub}}$ below the first layer.

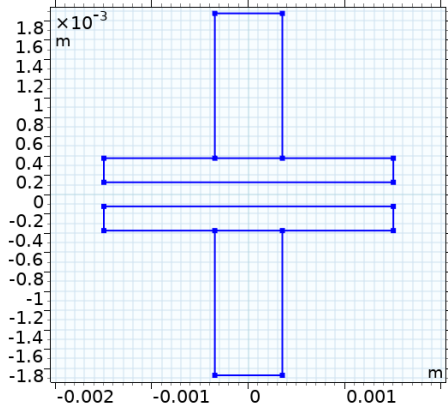


Figure 1. Representative PCB capacitor element, placed end-to-end to form each individual metawire in fig. 2. Length of the capacitor arms are adjusted to vary the impedance.

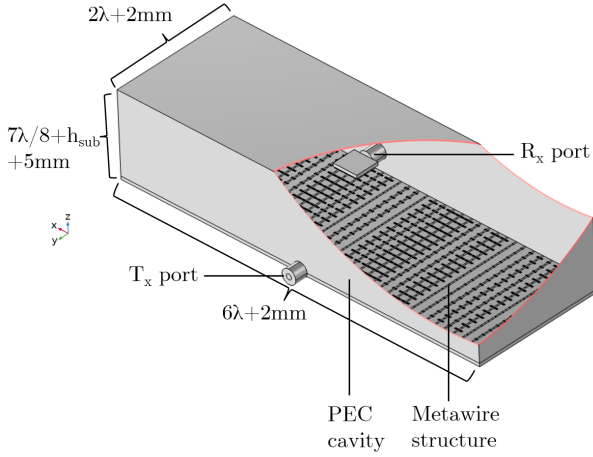


Figure 2. Cutaway model of optimized metawire structure enclosed in a PEC cavity. The entire setup depicted is enclosed in an airbox with perfectly matched layer (PML) boundary conditions (hidden in figure for clarity). Side y-z plane view in fig. 4. Optimized variables in table I.

The second layer supports a series-fed patch array (fig. 3a), which has the advantage of needing only a single input port and is easily implemented to excite an extended structure. A similar PCB capacitor metawire structure using an embedded line source for far-field beamforming has also been presented in [3].

A patch antenna (fig. 3b) is placed 1λ above the surface of the transmitting metawire structure, with a non-excited coaxial port. The enclosing PEC cavity measures $(7\lambda/8 + h_{\text{sub}} + 5 \text{ mm}) \times (2\lambda + 2 \text{ mm}) \times (6\lambda + 2 \text{ mm})$, i.e. $\sim 4.03 \times 6.20 \times 18.19 \text{ cm}$, a cavity size which adequately accommodates the metawire structure and the R_x patch antenna for demonstrative purposes in this work. The additional 1 mm gap on each side of the structure was added to provide a small tolerance for later fabrication.

III. FULL-WAVE SIMULATION

The underlying theoretical model of such metawire structures has been extensively discussed in previous works [3]–

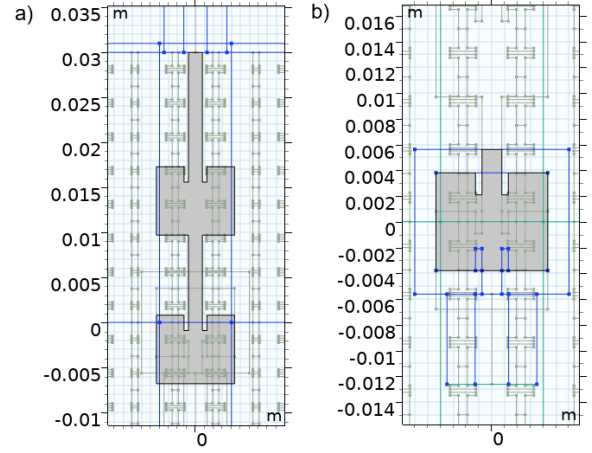


Figure 3. Dimensions of (a) series-fed patch used for exciting the metawire structure (fig. 2), and (b) resonant patch used as the R_x antenna.

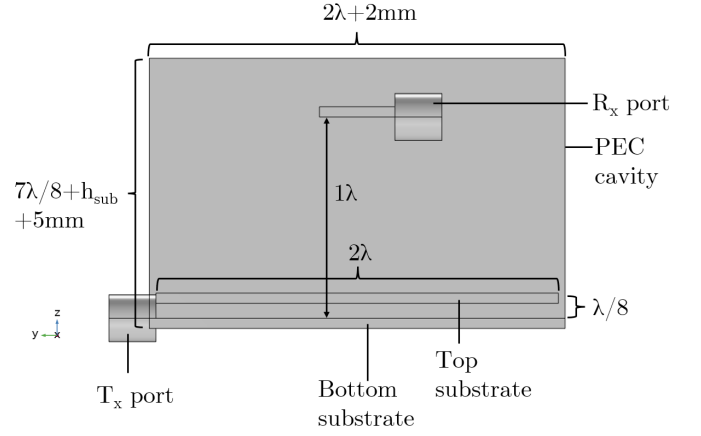


Figure 4. Cutaway side y-z plane view of metawire structure within cavity from fig. 2.

[5]. The metawire width is 0.7 mm and the thickness of capacitor arms is 0.25 mm (fig. 1), avoiding dimensions that are too fine for easy fabrication. Full-wave simulation is performed in COMSOL Multiphysics 6.2 using the RF module, with the “normal” physics-controlled mesh resolution setting. The coaxial T_x and R_x ports are explicitly modeled to best approximate the dimensions of a SMA cable connection (fig. 2, 4).

The derivative-free “Bound Optimization BY Quadratic Approximation” (BOBYQA) [6] method in COMSOL is used to maximize $\text{PTE} = P_{R_x}/P_{T_x}$, where P_{R_x} and P_{T_x} is the time-averaged Poynting flux integrated over the respective ports’ cross-sectional surface (i.e. $P_{\text{port}} = \frac{1}{2} \Re \left\{ \int_{S_{\text{port}}} \mathbf{E} \times \mathbf{H}^* \cdot \mathbf{n} dS \right\}$). For the 40-wire symmetry structure considered in this work (fig. 2) the optimization variables are the 20 capacitor widths for half of the full structure, each of which have an allowed range of [0.70, 4.25] mm corresponding to impedances of approximately [-100j, -30j]. The starting point is calculated using the optimization result for a 2-D

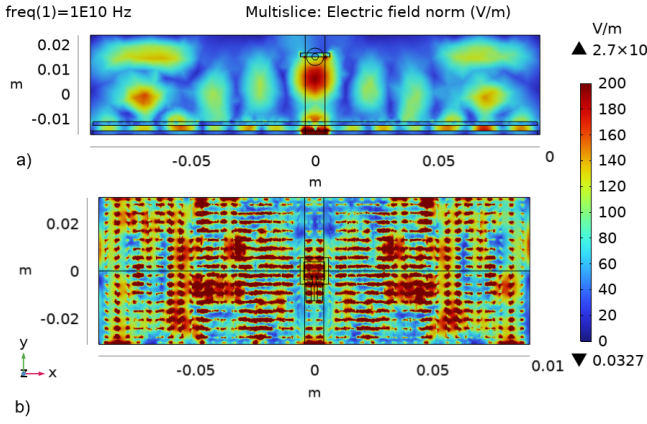


Figure 5. Total E field amplitude of optimized metawire structure (fig. 2), along (a) x-y plane and (b) 0.35 mm on the x-z plane above the metawire surface. The entirety of the metawire structure is excited from the series-fed patch. PTE is 85.0% (table II).

integral equation model, described in [3]. The final optimized capacitor widths from COMSOL are listed in table I; the capacitor width for each optimized metawire for half of the symmetric structure (w_1 to w_{10} , i.e. innermost to outermost) is specified.

Table I

WIDTHS OF OPTIMIZED CAPACITORS IN MM, FOR EACH METAWIRE IN FIG. 2.

Width, Capacitor	L (mm)	Width, Capacitor	L (mm)
w_1	2.0280	w_{11}	3.7939
w_2	0.73587	w_{12}	0.84857
w_3	4.0080	w_{13}	1.0993
w_4	2.6881	w_{14}	1.6201
w_5	4.2473	w_{15}	2.7555
w_6	3.9556	w_{16}	1.5912
w_7	4.2500	w_{17}	3.7090
w_8	1.7180	w_{18}	0.81824
w_9	1.8744	w_{19}	1.2914
w_{10}	4.2500	w_{20}	0.91701

A. Results and Comparison with Reference Benchmarks

Total E field from the optimized structure is shown in the fig. 5. The performance of the metawire structure (fig. 2) is compared with two benchmark examples in a cavity of the same dimensions: a scenario where the metawire metallization is removed (fig. 6), and a scenario with only two patch antennas separated by 1λ (fig. 7). The results are shown in table II, and are calculated for a scenario where the T_x port powers have been adjusted to the same amount.

Comparing the setup in fig. 2 to the setup in fig. 6, the presence of the metawire structure increases PTE while reducing the average E field intensity within the cavity from 321.7 V/m to 80.0 V/m. Furthermore, the internal cavity volume with $|E| \geq 100$ V/m is reduced from 94.2% to 24.7%, minimizing high fields in non-target regions (100 V/m is selected here as a representative threshold based on guidelines from [7], [8]). This indicates the metawires having a focusing effect towards

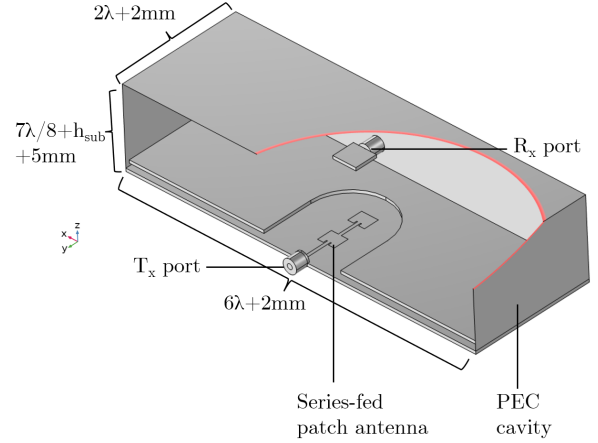


Figure 6. Cutaway model of reference setup with the metawire metallization removed. Another cutaway of the top substrate layer shows the series-fed patch. PTE is 19.0% (table II).

the R_x antenna, leading to the increased PTE and reduction of unwanted fields.

Comparing the setup in fig. 2 to the setup in fig. 7, the metawire structure shows substantial improvement in PTE as well as reduction of E field intensity in non-target regions. Although two antennas transmitting in a PEC cavity can be made to achieve 100% PTE, this requires using cavity resonance modes and positioning the antennas at locations of maximum field intensity; in real applications this may not always be possible, thus resulting in scenarios of diminished PTE as in fig. 7. Here the metawire structure is able to overcome this issue.

Table II

PTE AND E FIELD RESULTS. E FIELD DATA IS CALCULATED WITH INPUT PORT POWERS SCALED TO EQUAL THAT OF SETUP IN FIG. 2.

Setup	PTE, P_{R_x}/P_{T_x}	Average Cavity $ E $ (V/m)	Cavity Volume with $ E \geq 100$ V/m
Optimized Metawire Structure (fig. 2)	85.0%	80.0	24.7%
Benchmark: no metawires (fig. 6)	19.0%	321.7	94.2%
Benchmark: patch antenna pair (fig. 7)	52.2%	189.0	83.7%

IV. APPLICATIONS

The passive, low-profile metawire structure presented has the advantage of increasing PTE in enclosed cavity environments while reducing E field amplitude at unwanted locations, and not relying explicitly on resonance modes. Several potential applications can be considered:

A. Wireless Power in Existing Enclosed Environments

For applications where devices require wireless power in an existing enclosed conducting cavity (e.g. spacecraft/vehicle

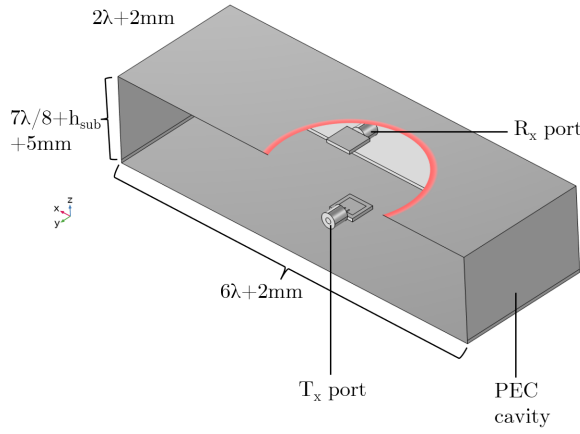


Figure 7. Cutaway model of reference setup with a pair of resonant 10 GHz patch antennas on 1.52 mm RO3003 substrate with coaxial feeds. Separation between the two antennas is 1λ . PTE is 52.2% (table II).

compartment), the proposed metawire structure is useful as it maximizes PTE while remaining unobtrusive and low-profile (fig. 2). In an ideal scenario placement of the T_x and R_x elements can exploit cavity resonances so that PTE is essentially 100%; however this is not always achievable as the cavity size and antenna placements may be constrained or non-ideal (fig. 7). Furthermore, obstructions could be present in the cavity, and movement of the R_x antenna away from field maxima points leads to rapid loss of PTE. The metawire structure overcomes these issues and can be adapted for more general scenarios. In addition, it also minimizes unwanted E fields while maintaining PTE and reduces interference with nearby equipment (table II).

B. Secure Charging and Medical Implant Charging

In the context of medical implants, the cavity and structure size can be scaled up to enclose the relevant region of the body for higher PTE charging; the working principle of the metawire structure can be redesigned for different more suitable frequencies, which may not necessarily be 10 GHz. The reduction of unwanted E fields is especially desirable, and the cavity prevents electromagnetic interference from the surroundings. The design can also be used for physically secure wireless charging of devices inside of a locked conducting box. In all cases, without explicitly relying on cavity resonances and strategic placement of antennas at field maxima locations, the design constraints are significantly lessened.

C. Power Harvesting

Power harvesting in enclosed environments can be achieved by essentially operating the system in reverse, i.e. reversing the roles of the T_x and R_x element in fig. 2. A conceptually similar super-absorbing surface is presented in [5] which can also be used for power harvesting applications, however it is optimized for incident plane waves, and assumes the entire surface of the planar metawire structure acts as the absorber by maximizing the dissipated power in the wires. For power harvesting the metawire structure presented in this work differs

from [5] in that instead of using the surface as an absorber, it redirect the incident power towards a specified R_x element (in this case the series-fed patch antenna). This has the advantage of simplifying the design implementation, without requiring a feed network or similar structures to collect the power.

V. CONCLUSION

This work presents the design of a passive cavity-enclosed PCB metawire structure for high efficiency WPT in a PEC cavity. Significant gains in PTE and reduction of unwanted fields are achieved compared to the reference scenarios using the same cavity setup. The results demonstrate the versatility of these metawire structures and show that, combined with their straightforward implementation, they are well suited to WPT for small electronics in applicable use cases. Relevant scenarios include charging small sensors in vehicle or spacecraft compartments where the enclosure cannot be resized, physically secure charging, and high-PTE charging with reduced off-target field intensity. Future work can extend the design to larger-scale cavities with the ability to accommodate larger targets, in addition to increasing PTE closer to 100%. Additional configurations, such as metawire structures with multiple embedded sources, could support off-angle power transfer and dynamic beamforming. Furthermore, such metawire structures may also accommodate dynamically reconfigurable or frequency-agile links, allowing even greater flexibility of use.

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